

YUXUAN ZHOU

Design Engineering | Product, Hardware, Embedded & AI-Enabled Systems

London, UK | +44 7818 079102 | zhouyuxuanpeter@outlook.com | [GitHub: Peter-zyx](https://github.com/Peter-zyx)

Seeking a 6-month industrial placement | Available Apr-Sep 2027

PROFILE

Design Engineering student at Imperial College London combining product design, mechanical engineering, embedded systems and AI-enabled software to develop end-to-end physical and digital products.

EDUCATION

Imperial College London - MEng Design Engineering

Expected Jun 2028

Relevant study: Data Science, Computing, Electronics, Embedded Systems, Mechanics, Thermofluids, Materials, CAD and Human-Centred Design.

PROFESSIONAL EXPERIENCE

Faust Network, Inc. - Product Design Intern

Sep 2026

- Designed the mechanical architecture and operating mechanism of an LED-ring subsystem for an AI hardware product, integrating components and defining hardware interaction and communication logic.
- Used ANSYS simulations to evaluate illumination and material behaviour, translating results into optical, material and structural decisions; built an interactive web app for internal testing and iteration.

Fudan University - AI & IoT Research Intern

Aug 2026

College of Intelligent Robotics and Advanced Manufacturing | Shanghai, China | [GitHub](https://github.com)

- Developed a multimodal smart medication-monitoring prototype integrating ESP32-S3, weight sensing, computer vision, on-device machine learning and a native iOS application.
- Built and deployed a hierarchical weight-classification pipeline that improved cross-session accuracy from 74% to 82%; developed MediaPipe + Core ML vision, BLE/Wi-Fi, camera streaming and multimodal decision logic for the end-to-end prototype.

Refuta - Analyst Intern

Jul-Aug 2025

Jiangsu, China

- Evaluated five warehouse software platforms against security, data-control, compliance and usability criteria; synthesised findings into a stakeholder recommendation and redesigned the corporate presentation template.

SELECTED PROJECTS

CP-APP: Our Little World - Product Design & AI-Assisted Development

Jul-Sep 2026

Two-person collaboration | [GitHub](https://github.com) | [Live Web](#)

- Designed the product concept, requirements, UI/UX, interaction model and feature set for a bilingual web and native iOS app for shared tasks, expenses, notes and memories.
- Directed and reviewed AI-assisted implementation across Next.js/React/TypeScript, SwiftUI and Supabase; iterated permissions, concurrency, notifications and location sharing, with 141 passing web/database tests.

Wulin: 2D Tactical Roguelike Web Game - Individual Project

May-Jun 2026

[GitHub](https://github.com) | [Play](#)

- Developed a playable grid-based tactical roguelike with turn combat, enemy pathfinding, terrain constraints and data-driven skills; separated game logic from content and wrote 32 automated tests for core systems.

Vehicle Aerodynamics & EV Battery Thermal Management - Team Project

Jan-Apr 2026

- Iterated an individual SUV concept in ANSYS CFD to Cd 0.286 and compared it with wind-tunnel testing of the team-selected model (Cd 0.31).
- Designed a dual-inlet aluminium cold plate; simulation predicted 42.5 °C average / 46.6 °C maximum temperature, 710.6 W heat removal and 15.1 Pa pressure drop.

Dancing Self-Balancing Segway - Team Project

Jan-Apr 2026

- Developed a self-balancing robot that recognised drum beats and mapped them to motion commands, using 125 ms MicroPython sampling, Hall-effect wheel-speed sensing and motor feedback.

TECHNICAL SKILLS

Product & CAD: Fusion 360, Rhino 3D, Blender; mechanical architecture, prototyping and 3D printing.

Simulation & Engineering: ANSYS Discovery and Workbench, CFD, FEA, optical/material studies and MATLAB.

Embedded & IoT: ESP32-S3, MicroPython, BLE/Wi-Fi, load cells, cameras, sensors and motors.

Software & AI: Python, Swift/SwiftUI, JavaScript/TypeScript, React/Next.js, Core ML, MediaPipe, Git/GitHub.

Design: Figma, Adobe Photoshop, Illustrator and InDesign; UI/UX, interaction design and product requirements.

ADDITIONAL INFORMATION

Languages: Mandarin Chinese (native), English (fluent; IELTS 8.0), Spanish (B1).

Interests: Robotics, computer vision, interactive systems, video games and travel.